HIGH LEVEL DESIGN

Banking Business Rules Engine (BRE)

Architecture, Components, Data Flow & Integration Design

Document	High Level Design — BRE Platform
Version	1.0
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Stack	Java 17 | Spring Boot 3.x | PostgreSQL | Redis | Kafka

1. Architecture Overview

The BRE is implemented as an independent microservice within the Lending Origination System ecosystem. It follows a layered architecture with clear separation between the Rule Management plane (configuration), the Rule Execution Engine (runtime), and the Persistence/Messaging plane (data and events).

1.1 Architecture Layers

LAYER 0 — External Consumers (LOS | LMS | Mobile App | CBS)
LAYER 1 — API Gateway / Load Balancer (REST + Kafka)
LAYER 2 — BRE Microservice | Rule Mgmt API | Execution API | Simulation API
LAYER 3 — Rule Engine Core | Rule Evaluator | Condition Parser | Outcome Resolver
LAYER 4 — Data | PostgreSQL (Rule Store) | Redis (Cache) | Kafka (Events) | S3 (Audit)

2. Component Architecture

2.1 BRE Microservice — Internal Components

Component	Responsibility	Key Classes / Beans
Rule Management API	CRUD for rules, versioning, status transitions	RuleController, RuleService, RuleVersionService
Rule Execution API	Synchronous rule evaluation endpoint	ExecutionController, RuleExecutionService
Rule Engine Core	Parses and evaluates rule conditions against input context	RuleEvaluator, ConditionParser, ExpressionEngine
Kafka Consumer	Async rule evaluation from Kafka events	BREEventConsumer, AsyncExecutionHandler
Simulation Engine	Test rules against mock or historical input	SimulationService, BacktestService
Rule Cache Manager	Cache active rules in Redis, invalidate on update	RuleCacheService, RedisCacheManager
Audit Logger	Persist every execution with input/output snapshot	AuditLogService, ExecutionLogRepository
Deviation Router	Aggregate deviations and determine escalation level	DeviationAggregator, EscalationResolver

3. Data Flow — Synchronous Rule Evaluation

3.1 Request Flow

Step	Actor	Action
1	LOS → BRE API	POST /api/v1/bre/evaluate with loan_app_id, product_type, rule_group, context payload
2	BRE API Layer	Validate request schema. Authenticate via JWT/API-key. Route to ExecutionService
3	ExecutionService	Load applicable ACTIVE rules for product_type + rule_group from Redis cache (fallback to DB)
4	RuleEvaluator	Evaluate each rule: parse conditions, compute expressions, resolve outcome (PASS/DEVIATION/FAIL)
5	DeviationAggregator	If any DEVIATION: aggregate all, determine highest escalation role
6	AuditLogger	Persist ExecutionLog with masked PII input snapshot and full output
7	Kafka Producer	Publish bre.evaluation.completed event for downstream consumers
8	BRE API → LOS	Return EvaluationResponse with rule results, overall status, deviation details

4. API Design — High Level

4.1 Rule Management APIs

Method	Endpoint	Auth Role	Description
POST	/api/v1/rules	RULE_AUTHOR	Create new rule in DRAFT status
PUT	/api/v1/rules/{id}	RULE_AUTHOR	Update rule (creates new version if ACTIVE)
POST	/api/v1/rules/{id}/activate	RULE_APPROVER	Activate approved rule for production
POST	/api/v1/rules/{id}/deactivate	RULE_APPROVER	Deactivate active rule
GET	/api/v1/rules/{id}/versions	RULE_VIEWER	Get full version history of a rule

4.2 Rule Execution APIs

Method	Endpoint	Auth Role	Description
POST	/api/v1/bre/evaluate	BRE_CONSUMER	Synchronous evaluation —

			returns results inline
POST	/api/v1/bre/evaluate/async	BRE_CONSUMER	Async evaluation — returns request ID; result via Kafka/webhook
POST	/api/v1/bre/simulate	RULE_AUTHOR	Simulate rule against test input without persisting
GET	/api/v1/bre/executions/{loanAppId}	BRE_VIEWER	Get full execution history for a loan application

5. Integration Architecture

5.1 Upstream Integrations

System	Protocol	Data Provided	SLA
LOS	REST / Kafka	Loan context, borrower, group data	< 200ms
Credit Bureau (CIBIL)	REST (sync)	Credit score, DPD history, active loans	< 3s (cached 24h)
VRM Service	REST	Village risk category	< 100ms (cached)
KYC Service	REST	Aadhaar/PAN verification status	< 500ms
Blacklist Service	REST	Debarment / defaulter flags	< 200ms

5.2 Downstream Integrations

System	Protocol	What BRE Sends
LOS Workflow	REST Response + Kafka	EvaluationResult: overall_status, rule_results[], deviation_role
Notification Service	Kafka event	bre.deviation.raised event with assignee and context
Audit / Compliance	Async DB + S3	Full execution log with input snapshot, rule versions evaluated
Reporting / MIS	Kafka / DB view	Rule execution summary aggregates

6. Technology Stack

Layer	Technology	Purpose

Application Framework	Java 17 + Spring Boot 3.x	Core microservice runtime
Rule Evaluation	Custom Expression Engine (SpEL)	Spring Expression Language for condition evaluation
Primary Database	PostgreSQL 15	Rule definitions, versions, execution logs
Cache	Redis 7 (Lettuce)	Active rule cache, bureau response cache
Messaging	Apache Kafka	Async rule evaluation, result events
API Security	Spring Security + JWT	Authentication + RBAC authorization
Service Discovery	Kubernetes / Eureka	Pod-level service registration
Monitoring	Micrometer + Prometheus + Grafana	Rule evaluation latency, throughput, error rate
Logging	SLF4J + ELK Stack	Centralized structured logging
CI/CD	Jenkins / GitHub Actions	Automated build, test, deploy pipeline

7. Deployment Architecture

7.1 Kubernetes Deployment Model

- BRE deployed as stateless pods with minimum 3 replicas for HA
- Horizontal Pod Autoscaler (HPA) scales based on CPU > 70% or request queue depth
- Redis deployed as Redis Sentinel (3 nodes) for HA cache
- PostgreSQL deployed with primary + 1 read replica (read replica for reporting queries)
- Kafka with minimum 3 brokers, replication factor 3 for all BRE topics
- All inter-service communication over internal cluster DNS — no public internet exposure

7.2 Environment Strategy

Environment	Purpose	Data	Rule Deployment
DEV	Active development and unit testing	Synthetic test data	Auto-deploy on commit
UAT	Business analyst rule simulation and acceptance	Anonymized production copy	Manual promotion from DEV
STAGING	Performance testing, integration validation	Production mirror	Manual promotion from UAT
PRODUCTION	Live rule evaluation	Live data	Approved rules only, dual approval required

8. Security Architecture

8.1 Authentication & Authorization

- All BRE APIs secured with JWT bearer tokens issued by the central IAM service
- RBAC enforced at method level using Spring Security @PreAuthorize annotations
- Service-to-service calls use mTLS with rotating certificates

Role	Permissions	Users
RULE_AUTHOR	Create, edit DRAFT/REVIEW rules, simulate	Credit Policy Analyst
RULE_APPROVER	Activate, deactivate rules, back-test	Credit Policy Head, Risk Officer
BRE_CONSUMER	Call evaluation APIs	LOS service, CBS service (system accounts)
BRE_VIEWER	Read-only: view rules, execution logs, reports	Audit team, Operations

8.2 Data Security

- PII fields (customer name, mobile, Aadhaar) masked in execution audit logs
- Rule execution inputs stored encrypted at rest in PostgreSQL (AES-256)
- Kafka topics for BRE events have ACL-based producer/consumer restrictions

9. Observability & Monitoring

Metric	Tool	Alert Threshold
Rule evaluation latency P99	Prometheus + Grafana	Alert if > 500ms for 5 consecutive minutes
Error rate	Prometheus	Alert if > 1% error rate over 2 minutes
Redis cache hit rate	Redis INFO + Grafana	Alert if < 80% (indicates DB pressure)
Kafka consumer lag	Kafka Manager / Grafana	Alert if lag > 1000 messages
Deviation rule rate	Custom Grafana dashboard	Monitor for unusual spikes — business alert

10. Disaster Recovery & Resilience

- Circuit breaker on all external integrations (Credit Bureau, KYC) using Resilience4J
- If bureau API is down: BRE proceeds with cached score or flags rule as PENDING_DATA
- Redis failure: BRE falls back to PostgreSQL for rule loading (< 50ms additional latency)

- Kafka failure: synchronous evaluations unaffected; async events queued in DB outbox pattern
- RTO (Recovery Time Objective): 5 minutes via Kubernetes pod restart
- RPO (Recovery Point Objective): 0 — all rule executions persisted synchronously before response
- PostgreSQL PITR (Point-in-Time Recovery) with 7-day backup retention